WEEK 4 – ASSIGNMENT

Name: Piyush Pankaj **Dit University**

Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

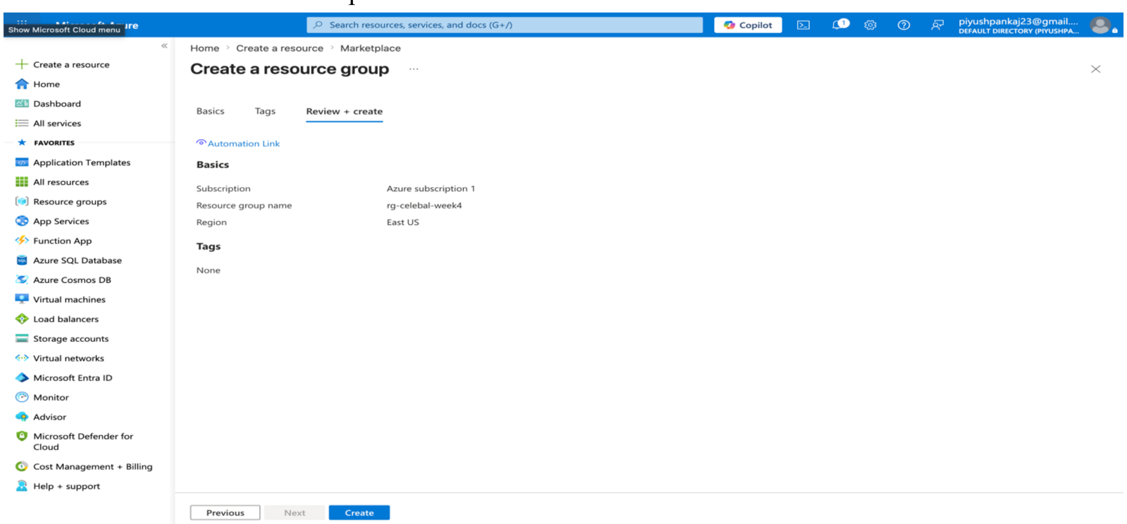
Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group



Resource group 'rg-celebal-week4' created (Review + Create)

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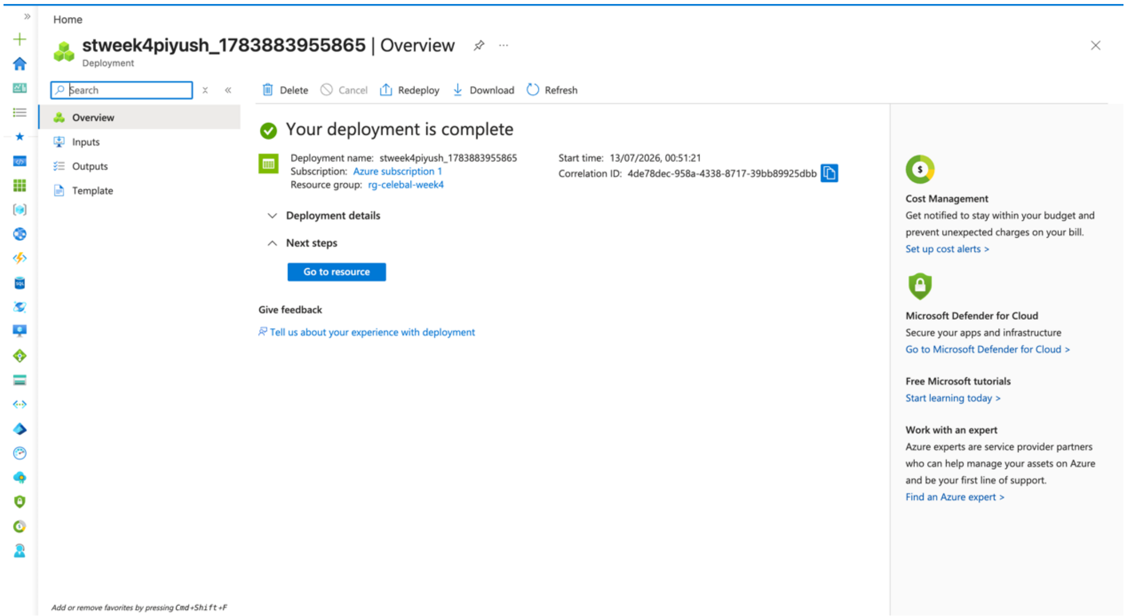


Resource group overview — showing Data Factory and Storage Account provisioned inside it

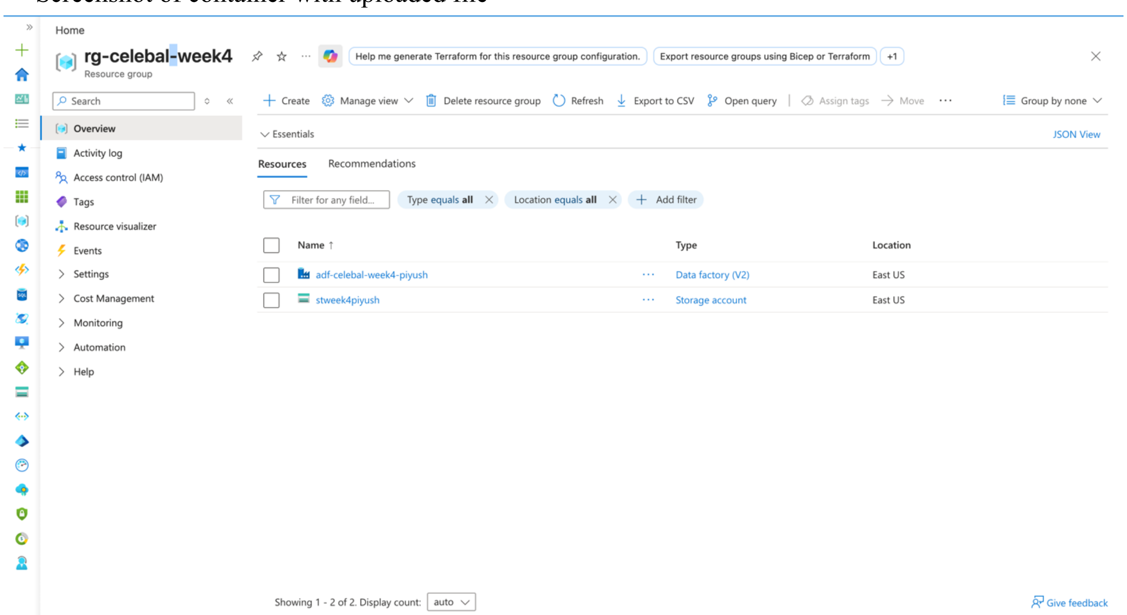
Task 2: Storage Setup

Do:

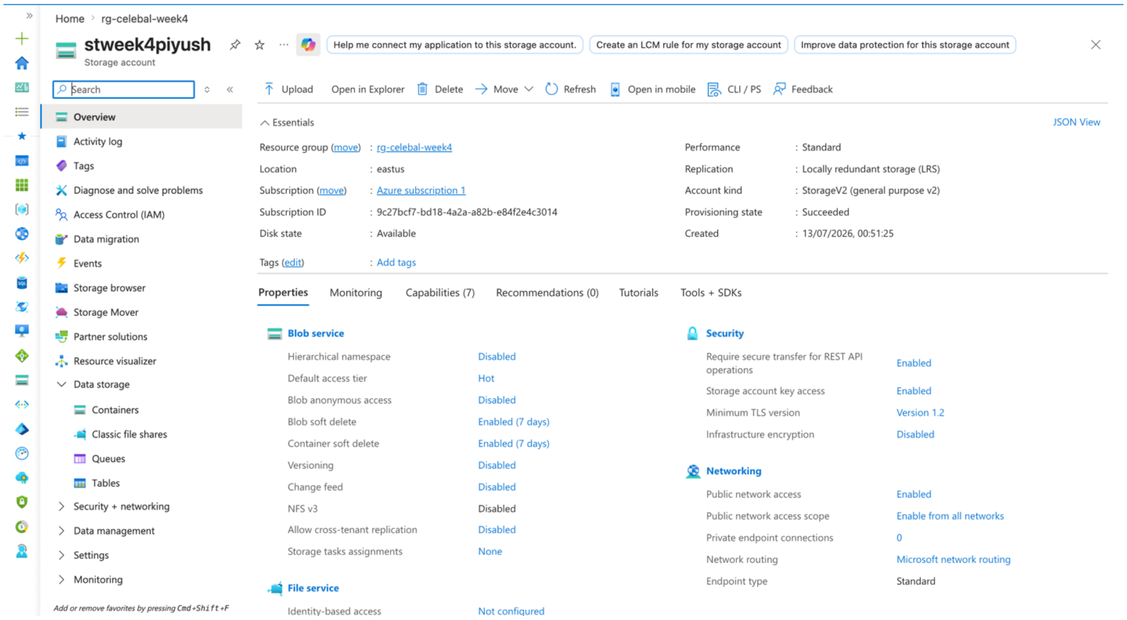
- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

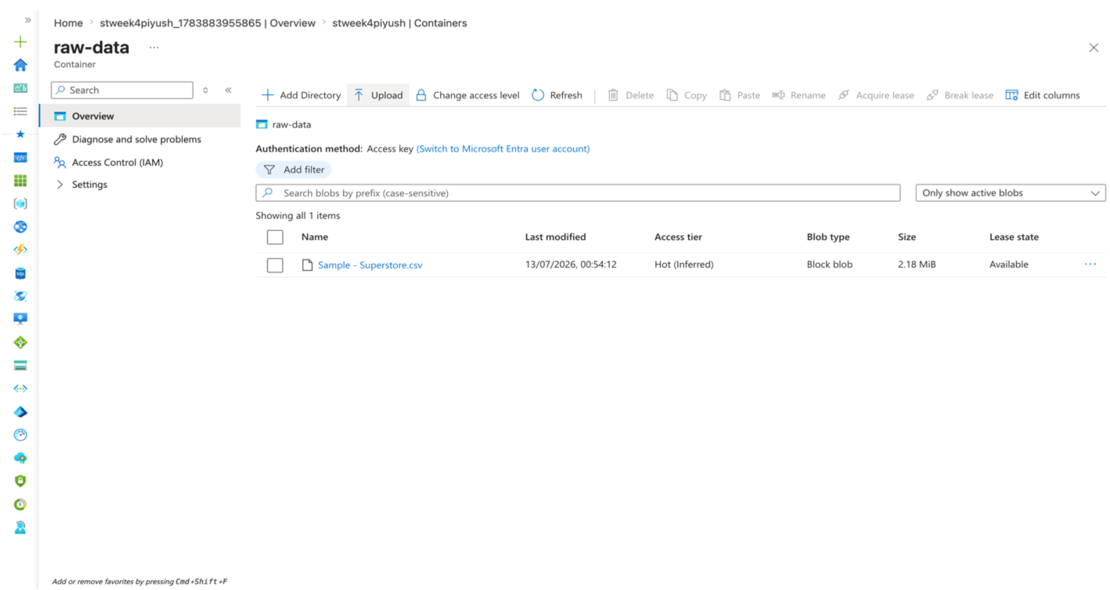
- Screenshot of container with uploaded file



Storage account deployment complete



Storage account overview / properties



'raw-data' container with 'Sample - Superstore.csv' uploaded

Task 3: ADF Basics

Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

Deliverable:

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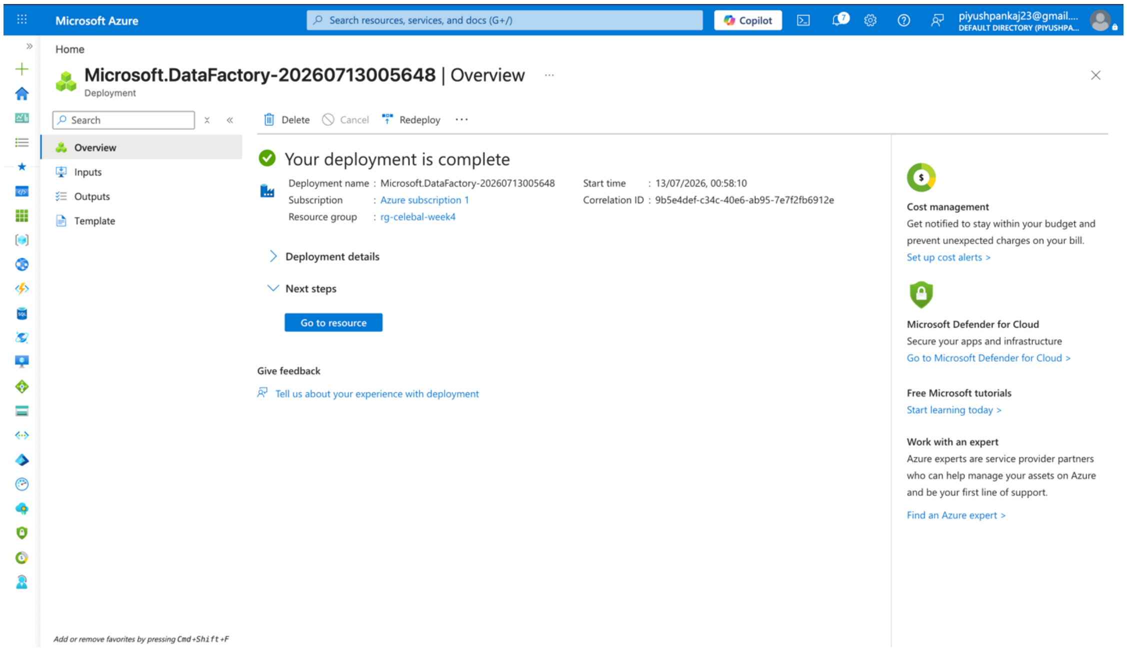
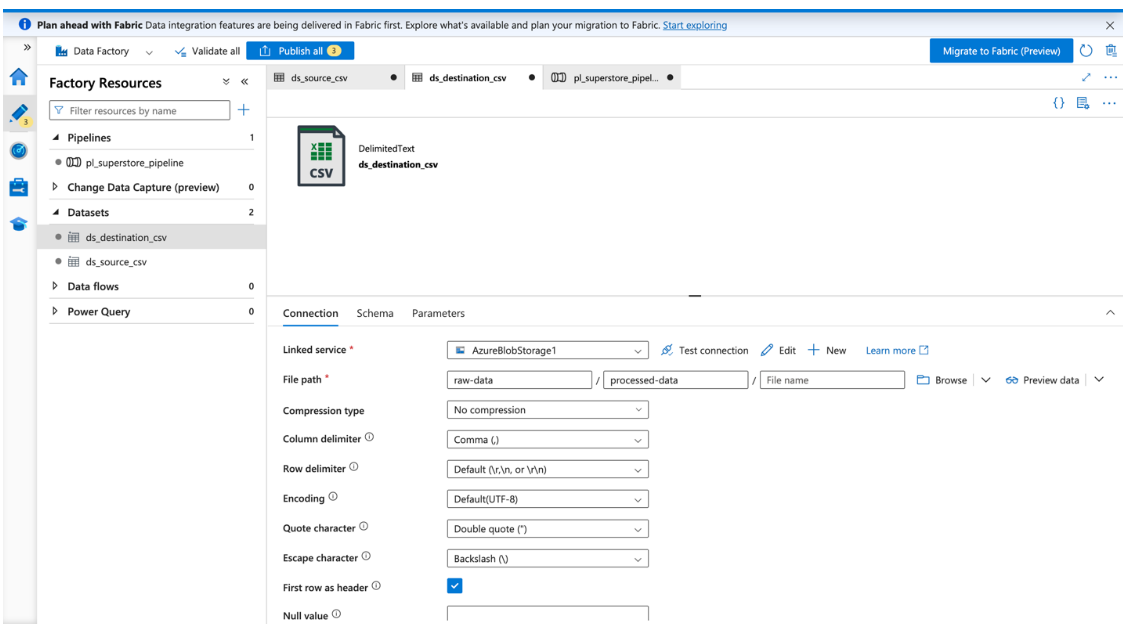
- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity



Azure Data Factory deployment complete



Linked Service 'AzureBlobStorage1' created

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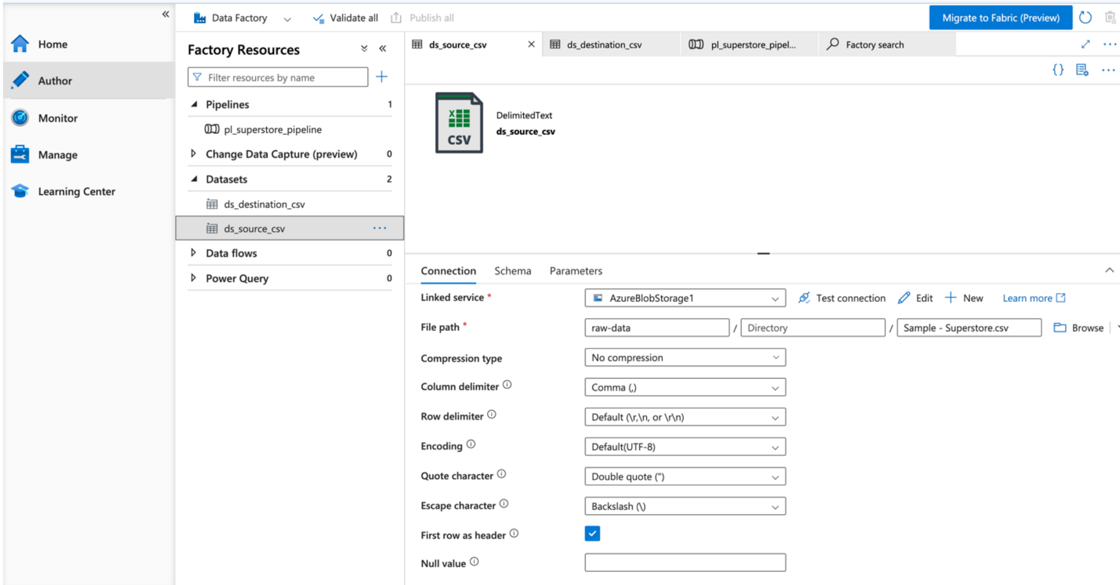
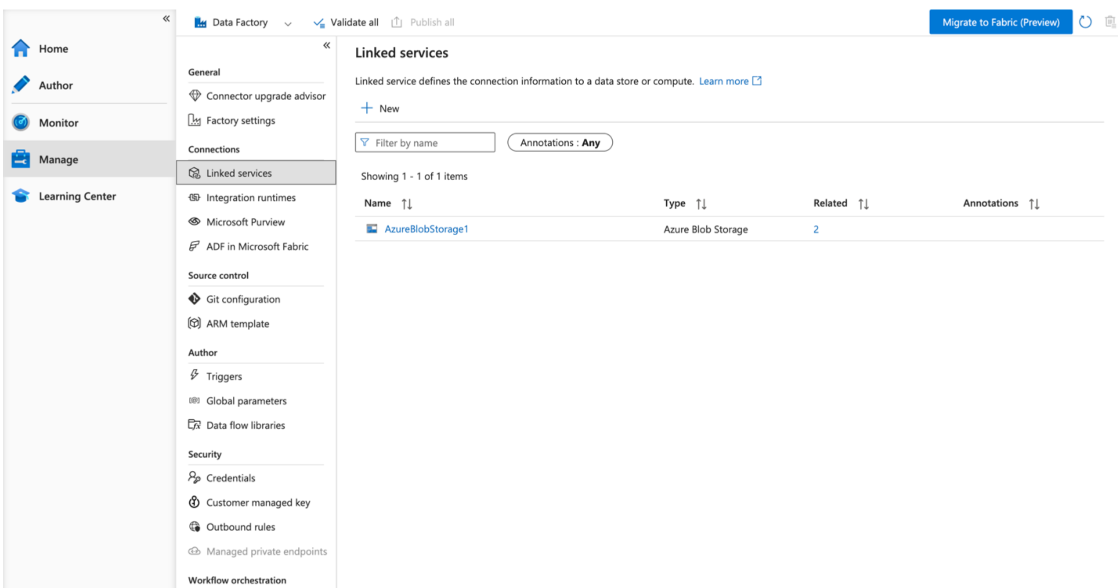


Source dataset 'ds_source_csv'



Destination dataset 'ds_destination_csv'

Input

Copy to clipboard


```
{
  "dataset": {
     "referenceName": "ds_source_csv",
     "type": "DatasetReference",
     "parameters": {}
  },
  "fieldList": [
     "itemName",
     "size",
     "lastModified"
  ],
  "storeSettings": {
     "type": "AzureBlobStorageReadSettings",
     "enablePartitionDiscovery": false
  },
  "formatSettings": {
     "type": "DelimitedTextReadSettings"
  }
}
```


Get Metadata activity — field list input (itemName, size, lastModified)

Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design

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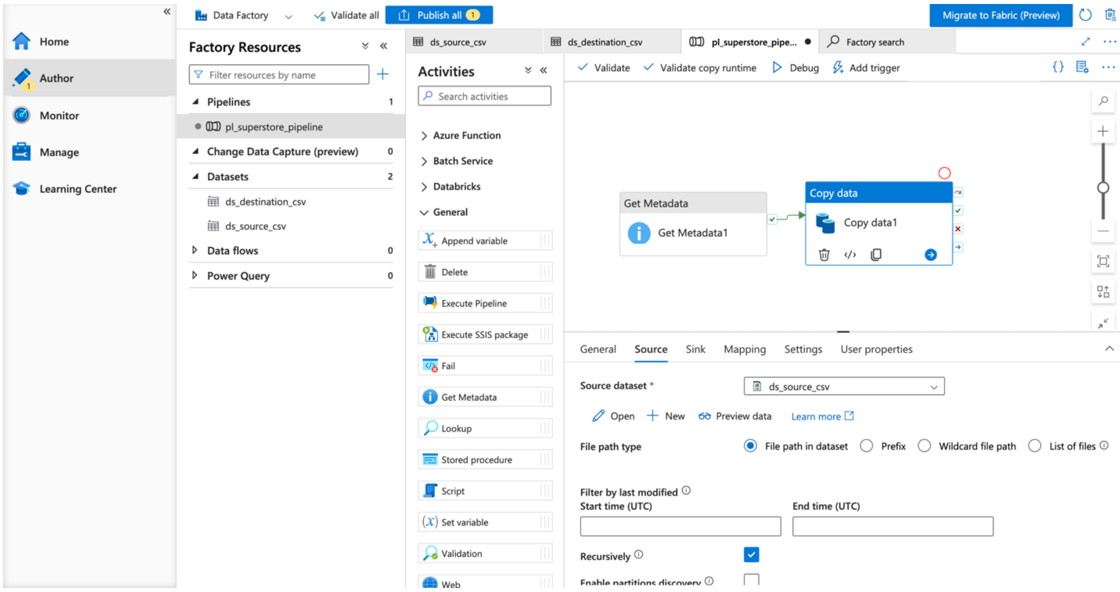
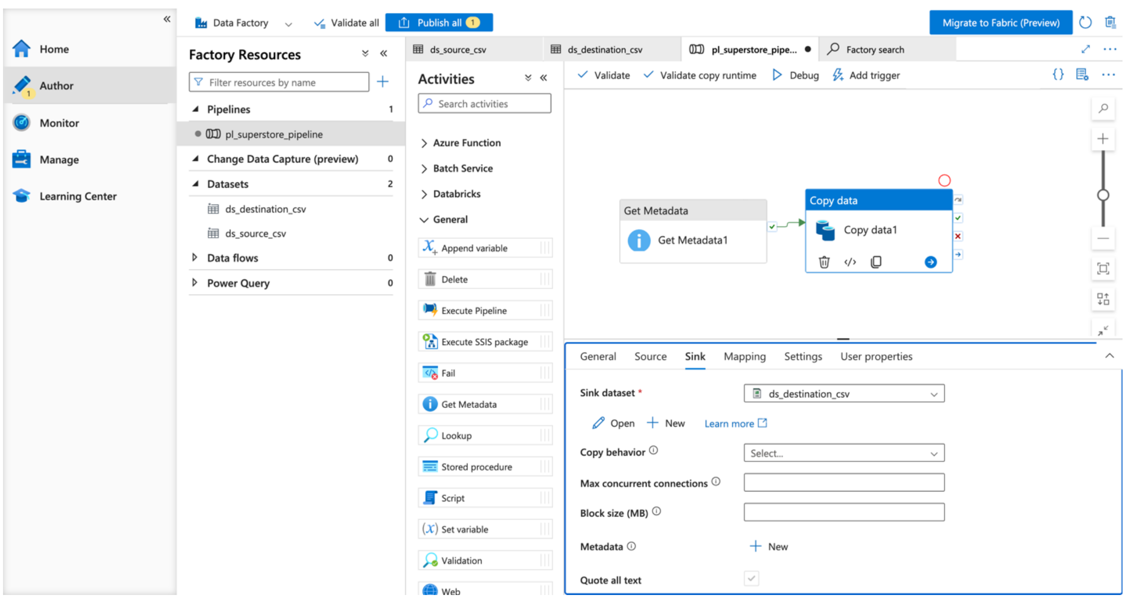


Pipeline design — Get Metadata connected to Copy Data (Source configuration)



Pipeline design — Copy Data Sink configuration

Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)

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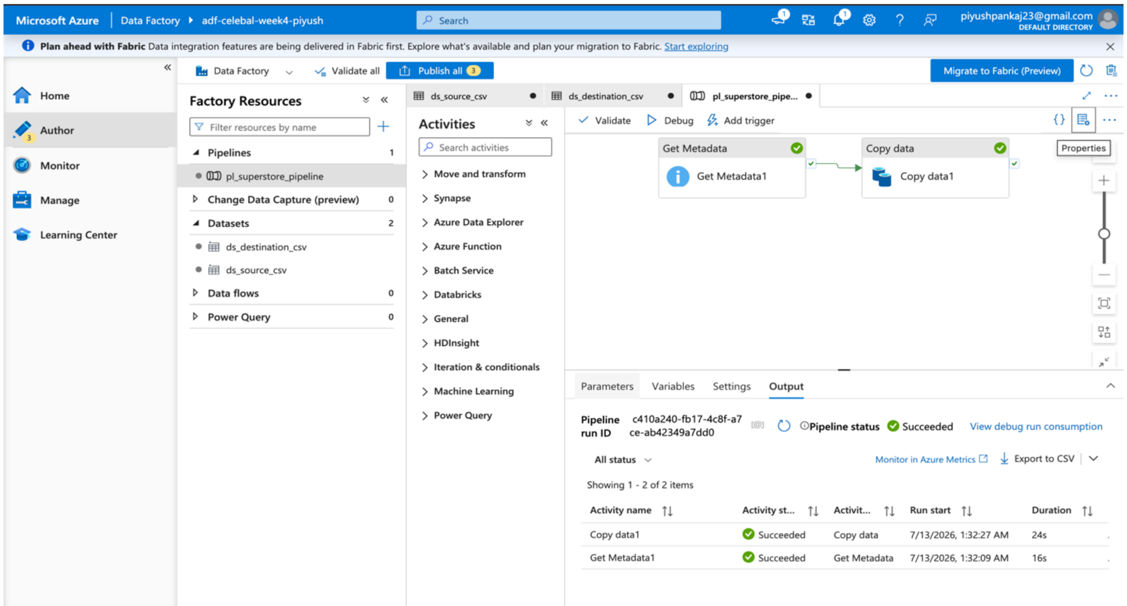


Pipeline run — both activities Succeeded

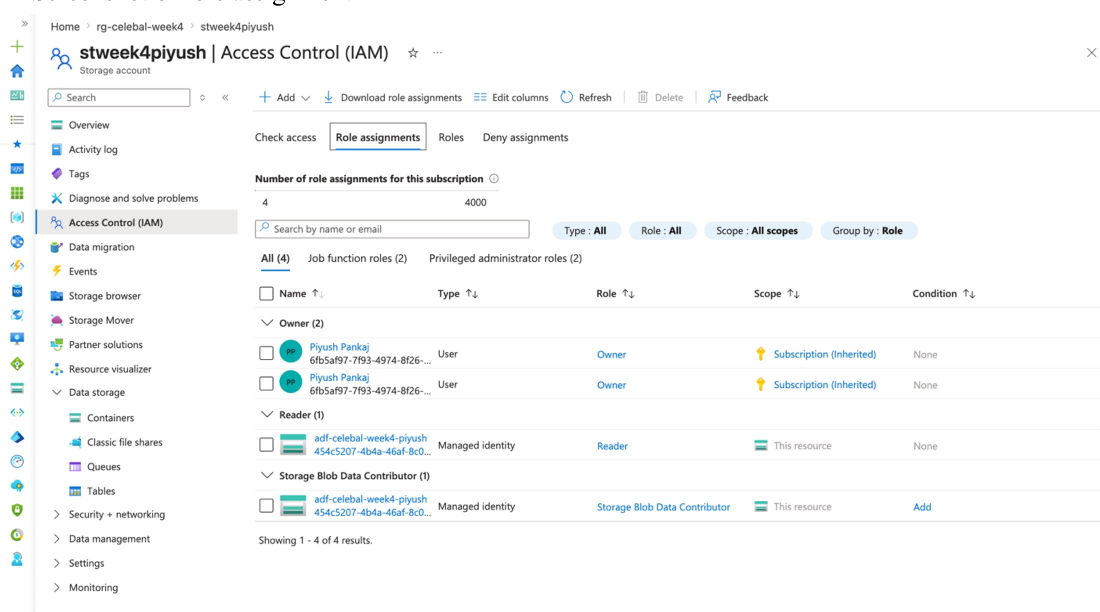
Task 6: IAM Roles

Do:

- Assign roles: Reader, Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment



Role assignments — ADF managed identity granted 'Reader' and 'Storage Blob Data Contributor' on the storage account

Mini Project (Final Task)

Problem Statement:

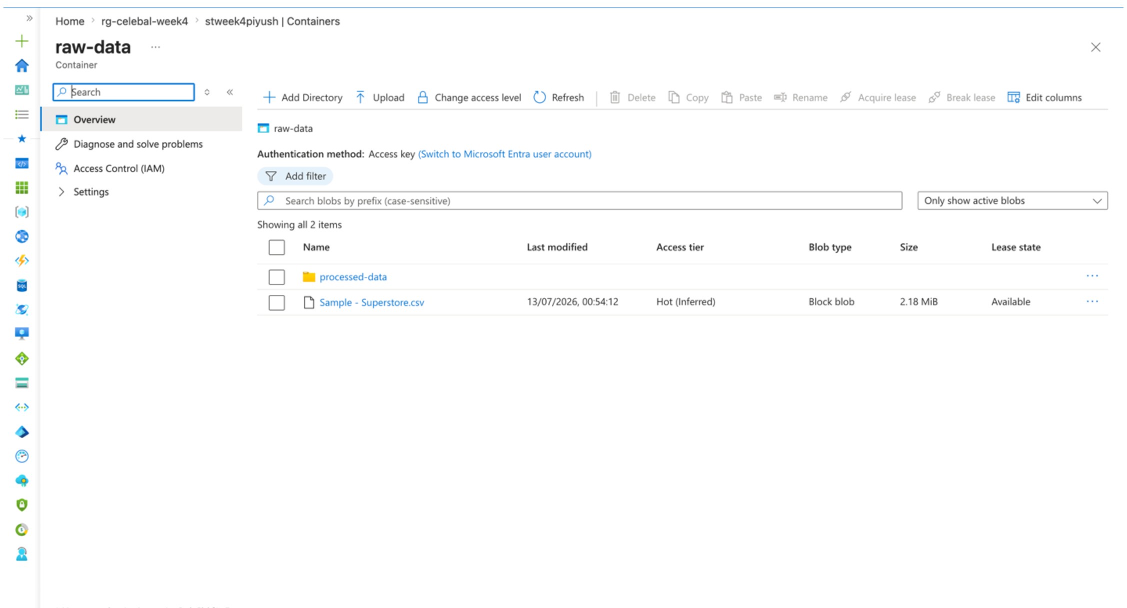
Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

Requirements:

- Source: CSV file in Blob
- Use: Linked Service + Dataset + Pipeline
- Process: Copy Data + Metadata check
- Destination: New file/location

Expected Output:

- Pipeline executed successfully
- Data copied to destination
- Metadata validated



'raw-data' container after pipeline run — 'processed-data' destination folder created alongside the source file

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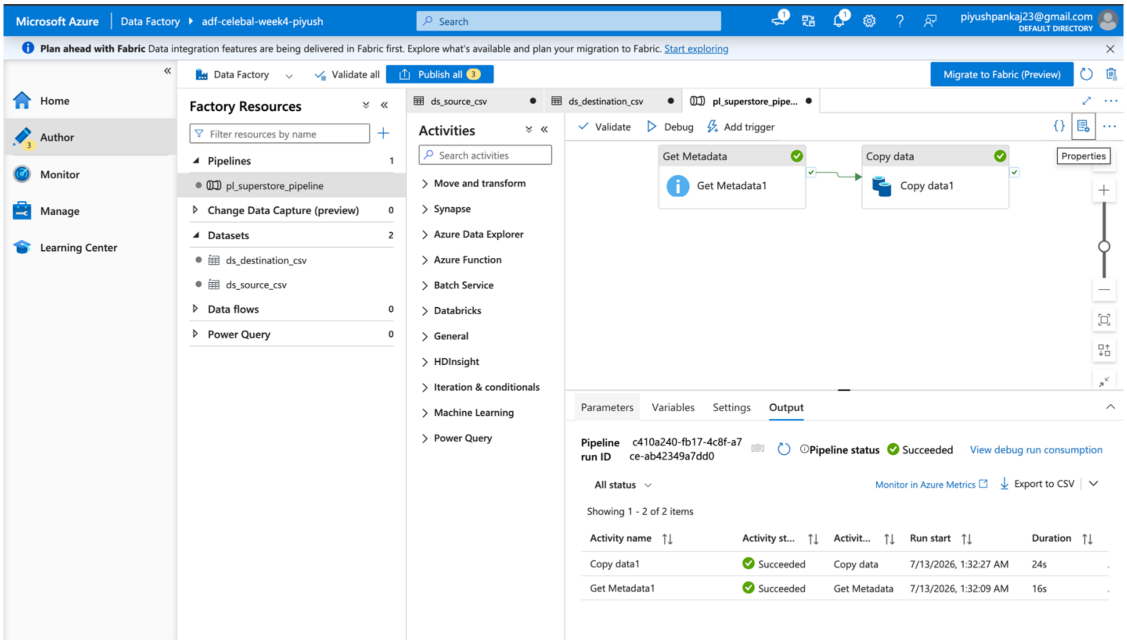

End-to-end pipeline (Get Metadata → Copy Data) executed successfully — metadata validated and data copied to destination

This mini project reuses the same end-to-end pipeline (pl_superstore_pipeline) built in Tasks 3–5. It reads 'Sample - Superstore.csv' from the 'raw-data' Blob container, validates its metadata (item name, size, last modified) via Get Metadata, and copies it to 'raw-data/processed-data/' via Copy Data. Both activities executed successfully, confirming the pipeline meets all stated requirements and expected outputs.